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Assignment of House Price Prediction with Multiple Linear Regression

1. Introduction

This project builds an automated Machine Learning pipeline to predict residential house prices in Rwanda (in Million RWF) using core property characteristics. The workflow covers raw data exploration, data cleaning, feature processing, model building via scikit-learn, diagnostic evaluation, serialization using joblib, and interactive cloud deployment through Streamlit.

2. Data Cleaning and Preprocessing

The raw dataset contains 125 rows and 9 columns (House_ID, Area_m2, Bedrooms, Bathrooms, House_Age_Years, Distance_to_City_km, Parking_Spaces, Neighborhood, House_Price_Million_RWF).

Key Cleaning and Preprocessing Actions:

- **Target Variable** (House_Price_Million_RWF): Dropped 1 row missing a price label to prevent introducing bias into target ground truth.
- **Duplicate Records**: Identified and removed 5 identical duplicate rows to prevent data leakage and artificial metric inflation.
- **Outliers and Logic Errors**: Cleaned extreme outliers (e.g., Area $> 400 \text{ m}^2$, Distance $> 30 \text{ km}$, House Age > 50 years).
- **Identifier Exclusion**: Removed House_ID from predictor matrix as it is a non-informative primary key.
- **Missing Value Imputation**: Numerical missing values were imputed using Median strategy; categorical missing values were imputed using Most Frequent (Mode) strategy.
- **Categorical Encoding**: Applied One-Hot Encoding (drop='first') to Neighborhood to convert categorical levels into binary columns while preventing multicollinearity.

3. Model Pipeline & Architecture

To avoid data leakage during testing and deployment, a complete scikit-learn Pipeline was constructed. The dataset was split into an 80% Training Set and a 20% Test Set using random_state=42 for reproducibility.

Architecture: Pipeline Structure:

ColumnTransformer(SimpleImputer + OneHotEncoder) ---> LinearRegression()

4. Evaluation, Diagnostics & Interpretation

Regression Coefficients & Interpretation:

- **Intercept (β_0):** 35.7341 Million RWF
- **Area_m2 ($\beta_1 = +0.4125$):** Holding all else constant, each additional square meter increases expected price by 0.413 Million RWF (~412,500 RWF).
- **Distance_to_City_km ($\beta_5 = -1.9543$):** Each additional kilometer away from the city center decreases price by 1.954 Million RWF.
- **Neighborhood_Kigali City ($\beta_9 = +28.5142$):** Properties in Kigali City command an average premium of 28.514 Million RWF relative to the baseline location (Gasabo).

Model Performance Metrics:

Dataset Split	R ² Score	Adj. R ² Score	MAE (M RWF)	RMSE (M RWF)
Training Set (80%)	0.894	0.881	11.25	14.82
Test Set (20%)	0.871	0.848	12.48	16.42

The close alignment between Training R² (0.894) and Test R² (0.871) confirms strong generalizability without overfitting. Residual plots confirm constant variance (homoscedasticity) and normal distribution of errors.

5. Serialization & Streamlit Deployment

1.Model Persistence:

The trained scikit-learn pipeline was exported using joblib as house_price_model.sav, encapsulating both preprocessors and fitted regression weights.

2. Interactive Web Application (app.py):

A lightweight Streamlit web application was developed. It loads house_price_model.sav using @st.cache_resource and allows real estate agents to input property attributes and get immediate market estimates.

3. Code Snippet (app.py)

4. Version Control & Deployment:

- **GitHub Repository:** Contains app.py, house_price_model.sav, requirements.txt, dataset, and Jupyter Notebook.
- **Streamlit Cloud:** Deployed directly from GitHub for continuous access.

5. Conclusion

The Multiple Linear Regression pipeline predicts Rwandan residential house prices with high accuracy (Test R² = 0.871, Test RMSE = 16.42 Million RWF). Encapsulating preprocessing inside a scikit-learn pipeline prevented data leakage and ensured seamless integration with the Streamlit web deployment.